



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2005

Yearly Examination

YEAR 11

Mathematics Extension

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 80 Marks

- Attempt Questions 1 - 4
- All questions are of equal value.

Examiner: *P. Bigelow*

Total marks – 80
Attempt Questions 1 - 4
All questions are of equal value

Answer each SECTION in a SEPARATE writing booklet.

Section A (Use a SEPARATE writing booklet)		Marks
Question 1 (20 marks)		
(a)	Find	
(i)	$\lim_{x \rightarrow 4} \frac{x^2 + x - 20}{x - 4}$	1
(ii)	$\lim_{x \rightarrow \infty} \frac{4 - x + 2x^2}{x^2 + 7x - 9}$	1
(b)	Find x in the following	
(i)	$8^x = 2^{x-4}$	1
(ii)	$\log_x 36 = 2$	1
(iii)	$\log_3 \frac{\sqrt{3}}{9} = x$	1
(c)	Differentiate the following	
(i)	$y = (7 - 2x)^6$	1
(ii)	$y = 7x^{-3} + 4x - 17$	1
(iii)	$f(x) = x\sqrt{x-2}$	2
(iv)	$f(x) = \frac{x+4}{x+5}$	2
(d)	Write down a quadratic equation with roots 4 and -7 .	2
(e)	If $f(x) = \sqrt{x^2 + 16}$ find $f'(3)$.	2
(f)	Find the equation of the tangent to $y = x^2 + 3x + 4$ at the point where $x = 0$.	2
(g)	For the parabola $(x + 2)^2 = 4y - 8$, find the:	
(i)	coordinates of the vertex;	1
(ii)	coordinates of the focus;	1
(iii)	equation of the directrix.	1

Section B (Use a SEPARATE writing booklet)**Marks****Question 2** (20 marks)

- (a) α, β and γ are the roots of $x^3 + 4x^2 + 8x + 16 = 0$.
Write down the values of:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1
- (iii) $\alpha\beta\gamma$ 1
- (iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 1
- (v) $\alpha^2 + \beta^2 + \gamma^2$ 1
- (b) Derive the equation of the normal to $x^2 = 4ay$ at $P(2as, as^2)$ 2
- (c) Write $\cos x - \sqrt{3}\sin x$ in the form $A\cos(x + \alpha)$ where $A > 0$
and $0 < \alpha < \frac{\pi}{2}$ 2
- (d) Solve:
- (i) $\frac{4}{|x+1|} < 3$ 2
- (ii) $\frac{3}{x-1} \geq \frac{2}{x}$ 2
- (e) Find the coordinates of the point which divides the interval AB ,
where A is $(-4, 0)$ and B is $(3, -7)$, *externally* in the ratio $3 : 2$ 2
- (f) (i) Show that -4 is a zero of $x^3 - 3x^2 - 18x + 40$. 1
- (ii) Hence solve $x^3 - 3x^2 - 18x + 40 = 0$ 2
- (g) Solve $\sin 2x = \cos x$ for $0 \leq x \leq 2\pi$ 2

Section C (Use a SEPARATE writing booklet)**Marks****Question 3** (20 marks)

- (a) Find the acute angle between the lines $2x - y - 7 = 0$ and $x - 3y + 3 = 0$. 2
- (b) Find the general solution of the equation $2 \cos \theta - \sqrt{3} = 0$ 2
- (c) (i) Write down the expansion of $\sin(A - B)$. 1
- (ii) Find the exact value of $\sin 15^\circ$. 2
- (d) The polynomial $x^3 + 2x^2 + ax + b$ has a factor of $x + 2$ and when divided by $x - 2$ the remainder is 12. Find the values of a and b . 3
- (e) (i) If $0 \leq \theta \leq \frac{\pi}{2}$ prove that $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$ 2
- (ii) Hence show that the exact value of $\tan \frac{\pi}{8}$ is $\sqrt{2} - 1$ 2
- (f) (i) Show that $\cot x + \tan x = 2 \operatorname{cosec} 2x$. 2
- (ii) Hence, or otherwise, prove that $\cot 15^\circ = 2 + \sqrt{3}$ 1
- (g) Show that if $2 \cos \theta = x + \frac{1}{x}$ then $2 \cos 3\theta = x^3 + \frac{1}{x^3}$. 3

You may use the identity $\cos 3A = 4 \cos^3 A - 3 \cos A$

Section D (Use a SEPARATE writing booklet)**Marks****Question 4** (20 marks)

- (a) Differentiate $f(x) = 3 - x + x^2$ using first principles. 2
- (b) Sketch the following without calculus:
- (i) $y = (x - 1)(x + 3)(x - 7)$ 2
- (ii) $y = (x - 2)^2(x + 3)$ 2
- (c) $P(2p, p^2)$ and $Q(2q, q^2)$ are points on the parabola $x^2 = 4y$.
- (i) Show that the equation of the chord joining P and Q is given by $y = \left(\frac{p+q}{2}\right)x - pq$ 2
- (ii) Find the coordinates of the midpoint, Z , of PQ . 2
- (iii) If the chord, joining P and Q , passes through $(0, 2)$ find the locus of Z . 3
- (d) (i) θ and ϕ are acute angles such that $\cos \theta = \frac{3}{5}$ and $\sin \phi = \frac{1}{\sqrt{5}}$. 2
Without finding the size of either angle, show that $\theta = 2\phi$.
- (ii) Using the result above, find the exact value of $\sin 3\phi$ 3
- (e) Given that $\log_b a = 2$ and $\log_c b = 3$, find the value of $\log_a c$. 2

End of paper



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Sample Solutions

Section	Marker
A	DM
B	FN
C	EC
D	AF

SECTION A Year 11 Extension Mathematics

Q1

$$a) i) \lim_{x \rightarrow 4} \frac{x^2 + x - 20}{x - 4} = \lim_{x \rightarrow 4} \frac{(x-4)(x+5)}{x-4}$$

$$= \lim_{x \rightarrow 4} x + 5$$

$$= 9.$$

(1)

$$ii) \lim_{x \rightarrow \infty} \frac{4 - x + 2x^2}{x^2 + 7x - 9} = \lim_{x \rightarrow \infty} \frac{4 - x + 2x^2}{x^2 + 7x - 9} \times \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{4}{x^2} - \frac{1}{x} + 2}{1 + \frac{7}{x} - \frac{9}{x^2}}$$

$$= \frac{0 - 0 + 2}{1 + 0 - 0}$$

$$= 2.$$

(1)

$$b) 8^x = 2^{x-4}$$

$$\log_2 8^x = \log_2 2^{x-4}$$

$$x \log_2 8 = (x-4) \log_2 2$$

(1)

$$3x \log_2 2 = x - 4$$

$$3x = x - 4$$

$$2x = -4$$

$$x = -2.$$

$$\text{ii) } \log_x 36 = 2$$

$$36 = x^2$$

①

$$x = 6.$$

$$\text{ii) } \log_3 \frac{\sqrt{3}}{9} = x$$

$$\frac{\sqrt{3}}{9} = 3^x$$

①

$$3^{\frac{1}{2}} \div 3^2 = 3^x$$

$$3^{-\frac{3}{2}} = 3^x$$

$$x = -\frac{3}{2}.$$

$$\text{c) i) } y = (7-2x)^6$$

Chain rule

$$\frac{dy}{dx} = 6(7-2x)^5(-2)$$

①

$$= -12(7-2x)^5$$

$$\text{ii) } y = 7x^{-3} + 4x - 17$$

$$\frac{dy}{dx} = -21x^{-4} + 4$$

①

$$\text{iii) } f(x) = x\sqrt{x-2}$$

Chain and product rule.

$$f'(x) = \sqrt{x-2} + \frac{x}{2}(x-2)^{-\frac{1}{2}}$$

②

$$= \sqrt{x-2} + \frac{x}{2\sqrt{x-2}}$$

$$= \sqrt{x-2} \left(\frac{3x-4}{2x-4} \right)$$

$$1v) f(x) = \frac{x+4}{x+5}$$

Quotient rule.

$$f'(x) = \frac{(x+5) - (x+4)}{(x+5)^2}$$

(2)

$$= \frac{1}{(x+5)^2}$$

$$d) (x-4)(x+7) = 0.$$

$$x^2 - 4x + 7x - 28 = 0$$

$$x^2 + 3x - 28 = 0.$$

(2)

$$e) f(x) = \sqrt{x^2 + 16}$$

Chain rule

$$f'(x) = \frac{1}{2} (x^2 + 16)^{-\frac{1}{2}} (2x)$$

(2)

$$= \frac{x}{\sqrt{x^2 + 16}}$$

$$f'(3) = \frac{3}{\sqrt{9 + 16}}$$

$$= \frac{3}{5}$$

$$f) y = x^2 + 3x + 4$$

At $x=0$

$$y = 0^2 + 3(0) + 4$$

$$y = 4$$

(0, 4), is point of contact.

$$\frac{dy}{dx} = 2x + 3$$

At (0, 4)

$$\frac{dy}{dx} = 2(0) + 3$$

$$= 3$$

Eqn of tangent $m=3$, (0, 4).

$$y - 4 = 3(x - 0)$$

$$y = 3x + 4.$$

(2)

q 1) $(x+2)^2 = 4y - 8$

$$(x+2)^2 = 4(y-2)$$

①

Vertex $(-2, 2)$

ii) focal length $= 1$

①

∴ focus $(-2, 3)$

iii) Directrix $y = 1$

①

QUESTION 2

(a) $x^3 + 4x^2 + 8x + 16 = 0$

(i) $\alpha + \beta + \gamma = -\frac{b}{a} = -4$

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = 8$

(iii) $\alpha\beta\gamma = -\frac{d}{a} = -16$

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma} = \frac{8}{-16} = -\frac{1}{2}$

(v) $(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) = 16 - 2 \times 8 = 0$

(b) $x^2 = 4ay \quad (2as, as^2)$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a} = \frac{x}{2a} \text{ (grad. of tangent)}$$

$$y' = \frac{2as}{2a} = s \text{ (} x = 2as \text{)}$$

$$\text{grad. of normal} = -\frac{1}{s}$$

$$\text{eqn. of normal is } y - as^2 = -\frac{1}{s}(x - 2as)$$

$$x + sy = as^3 + 2as$$

(c) $\cos x - \sqrt{3} \sin x = A \cos(x + \alpha)$

$$\cos x - \sqrt{3} \sin x = A \cos x \cos \alpha - A \sin x \sin \alpha$$

$$\text{equating coefficients } A \cos \alpha = 1, A \sin \alpha = \sqrt{3}$$

$$\tan \alpha = \sqrt{3} \quad \alpha = \frac{\pi}{3} \quad A = 2$$

$$\cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$$

d) (i) $\frac{4}{|x+1|} < 3$

$$4 < 3|x+1| \quad (|x+1| > 0)$$

$$4 < 3(x+1) \text{ or } 4 < 3(-x-1)$$

$$3x+3 > 4 \text{ or } -3x-3 > 4$$

$$3x > 1 \text{ or } -3x > 7$$

$$x > \frac{1}{3} \text{ or } x < -\frac{7}{3}$$

(d) (ii) $\frac{3}{x-1} > \frac{2}{x} \quad x \neq 1, x \neq 0$

$$\left[\text{multiply by } (x-1)^2 x^2 \right]$$

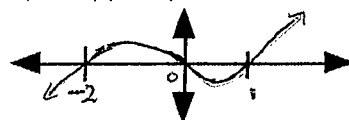
$$3x^2(x-1) \geq 2x(x-1)^2$$

$$3x^3 - 3x^2 \geq 2x^3 - 4x^2 + 2x$$

$$x^3 + x^2 - 2x \geq 0$$

$$x(x^2 + x - 2) \geq 0$$

$$x(x+2)(x-1) \geq 0$$



$$-2 \leq x < 0 \text{ or } x \geq 1$$

(e) $m:n = 3:-2 \quad (-4, 0) \quad (3, -7)$

$$x = \frac{mx_2 + nx_1}{m+n} \quad y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{3 \times 3 - 2 \times -4}{3-2} \quad y = \frac{3 \times -7 - 2 \times 0}{3-2}$$

$$(17, -21)$$

(f) (i) $P(-4) = (-4)^3 - 3(-4)^2 - 18(-4) + 40 = 0$
or

$$x + 4 \overline{\begin{array}{r} x^2 - 7x + 10 \\ x^3 - 3x^2 - 18x + 40 \end{array}}$$

$$\begin{array}{r} x^3 + 4x^2 \\ -7x^2 - 18x \\ -7x^2 - 28x \end{array}$$

$$10x + 40$$

$$10x + 40$$

$$0 + 0$$

$$\text{remainder} = 0 \quad \therefore -4 \text{ is a zero of } P(x)$$

(ii) $x^3 - 3x^2 - 18x + 40 = (x+4)(x-5)(x-2) = 0$
 $x = -4, 2, 5$

(g) $\sin 2x - \cos x = 0$
 $2 \sin x \cos x - \cos x = 0$
 $\cos x (2 \sin x - 1) = 0$

$$\cos x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \text{or} \quad x = \frac{\pi}{6}, \frac{5\pi}{6}$$

Section (c) — Solutions

$$(a) \quad 2x - y - 7 = 0 \quad \text{--- (1)}$$

$$x - 3y + 3 = 0 \quad \text{--- (2)}$$

$$y = 2x + 7, \quad m_1 = 2$$

$$y = \frac{x}{3} + 1, \quad m_2 = \frac{1}{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - \frac{1}{3}}{1 + \frac{2}{3}} \right|$$

$$= 1$$

$$\therefore \theta = 45^\circ$$

$$(c) \quad 2 \cot \theta = \frac{\sqrt{3}}{2}$$

$$\cot \theta = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = 2n\pi \pm \frac{\pi}{6}$$

$$(c) \quad \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$(i) \quad \sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$(ii) = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2}(\sqrt{3}-1)}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

$$(d) \quad x^3 + 2x^2 + ax + b = P(x)$$

$$P(-2) = 0$$

$$-8 + 8 - 2a + b = 0$$

$$\therefore b = 2a \quad \text{--- (1)}$$

$$P(2) = 12$$

$$8 + 8 + 2a + b = 12$$

$$2a + b = -4 \quad \text{--- (2)}$$

$$\text{Subst. (1) into (2)}$$

$$4a = -4 \quad \therefore a = -1, \quad b = -2$$

$$(e) \quad \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$$

$$(i) \quad \sqrt{\frac{(\sin^2 \theta + \cos^2 \theta) - (\cos^2 \theta - \sin^2 \theta)}{2 \cos^2 \theta}}$$

$$= \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} = \tan \theta$$

$$\therefore \tan \frac{\pi}{8} = \sqrt{\frac{1 - \cos \frac{\pi}{4}}{1 + \cos \frac{\pi}{4}}}$$

$$= \sqrt{\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}}$$

$$= \sqrt{\frac{\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right) \times \frac{\sqrt{2}-1}{\sqrt{2}-1}}{(\sqrt{2}-1)^2}} = \sqrt{2} - 1$$

$$(f)$$

$$(i) \quad \cot x + \frac{1}{\cot x}$$

$$= \frac{\cot^2 x + 1}{\cot x}$$

$$= \frac{\sec^2 x}{\cot x}$$

$$= \frac{1}{\sin^2 x} \times \frac{\sin x}{\cos x}$$

$$= \frac{2}{2 \sin x \cos x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

$$(ii) \quad \cot 15^\circ + \frac{1}{\cot 15^\circ} = \frac{2}{\sin 30^\circ}$$

$$\therefore \cot 15^\circ + \frac{1}{\cot 15^\circ} = 4$$

$$\cot^2 15^\circ - 4 \cot 15^\circ + 1 = 0$$

$$\text{Let } m = \cot 15^\circ$$

$$m^2 - 4m + 1 = 0$$

$$m = \frac{4 \pm \sqrt{16-4}}{2}$$

$$= 2 \pm \sqrt{3}$$

$$\therefore \cot 15^\circ = 2 + \sqrt{3}$$

(g)

$$\text{If } 2\cos\theta = x + \frac{1}{x},$$

$$8\cos^3\theta = \left(x + \frac{1}{x}\right)^3$$

$$= x^3 + 3x + \frac{3}{x} + \frac{1}{x^3}$$

$$= \left(x^3 + \frac{1}{x^3}\right) + 3\left(x + \frac{1}{x}\right) \text{ ————— (1)}$$

$$\text{but } \cos^3 A = 4\cos^3 A - 3\cos A$$

$$\therefore 2\cos^3 A = 8\cos^3 A - 6\cos A$$

From (1)

$$x^3 + \frac{1}{x^3} = 8\cos^3\theta - 6\cos\theta$$

$$\therefore x^3 + \frac{1}{x^3} = 2\cos^3\theta \text{ ————— (2)}$$

$$4.a. \quad f(x) = 3 - x + x^2$$

$$f(x+h) = 3 - (x+h) + (x+h)^2$$

$$= 3 - x - h + x^2 + 2xh + h^2$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3} - \cancel{x} - h + \cancel{x^2} + 2xh + h^2 - (\cancel{3} - \cancel{x} + \cancel{x^2})}{h}$$

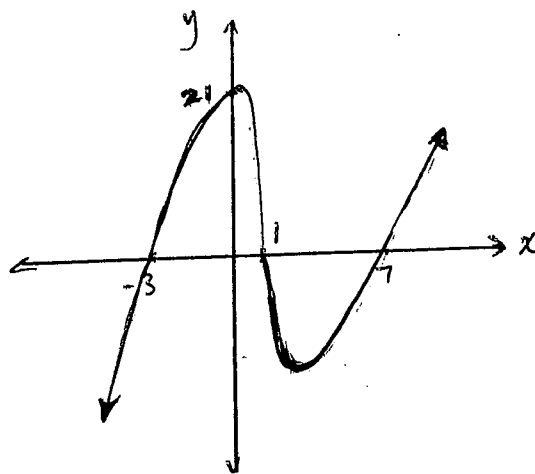
$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(-1 + 2x + h)}{\cancel{h}}$$

$$= -1 + 2x$$

$$b.i. \quad y = (x-1)(x+3)(x-7)$$

$$\text{when } x=0, y=21$$

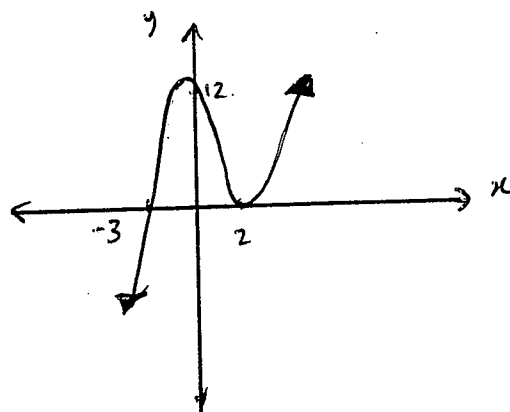
$$y=0, x=1, -3, 7$$



$$ii. \quad y = (x-2)^2(x+3)$$

$$\text{when } x=0, y=12$$

$$y=0, x=2, -3$$



$$c.i. \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{p^2 - q^2}{2p - 2q}$$

$$= \frac{(p-q)(p+q)}{2(p-q)}$$

$$= \frac{p+q}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - p^2 = \frac{p+q}{2} (x - 2p)$$

$$y = \left(\frac{p+q}{2}\right)x - p^2 - pq + p^2$$

$$y = \left(\frac{p+q}{2}\right)x - pq$$

$$\text{ii. } Z = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$= \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2}\right)$$

$$= (p+q, \frac{p^2+q^2}{2})$$

$$\text{iii. } (0, 2) \text{ satisfies } y = \left(\frac{p+q}{2}\right)x - pq$$

$$2 = \left(\frac{p+q}{2}\right) \cdot 0 - pq$$

$$-pq = 2$$

$$pq = -2 \quad \text{--- (1)}$$

$$\text{from (ii) } x = p+q \quad \text{--- (2)}$$

$$y = \frac{p^2+q^2}{2} \quad \text{--- (3)}$$

rearrange (3)

$$y = \frac{(p+q)^2 - 2pq}{2} \quad \text{--- (3a)}$$

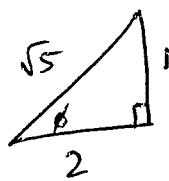
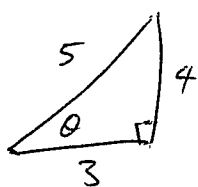
substitute (1) & (2) into (3a)

$$y = \frac{x^2 - 2(-2)}{2}$$

$$y = \frac{x^2 + 4}{2}$$

d.i. θ & ϕ are acute angles

$$\cos \theta = \frac{3}{5}, \quad \sin \phi = \frac{1}{\sqrt{5}}$$



Show that $\theta = 2\phi$

$$\sin \theta = \sin 2\phi$$

$$\text{LHS} = \frac{4}{5}$$

$$\begin{aligned} \text{RHS} &= 2 \sin \phi \cos \phi \\ &= 2 \cdot \frac{1}{\sqrt{5}} \cdot \frac{2}{\sqrt{5}} \\ &= \frac{4}{5} \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

$$\sin \theta = \sin 2\phi$$

$$\therefore \theta = 2\phi$$

$$\begin{aligned} \text{ii. } \sin 3\phi &= \sin(2\phi + \phi) \\ &= \sin(\theta + \phi) \\ &= \sin \theta \cos \phi + \cos \theta \sin \phi \\ &= \frac{4}{5} \cdot \frac{2}{\sqrt{5}} + \frac{3}{5} \cdot \frac{1}{\sqrt{5}} \\ &= \frac{8}{5\sqrt{5}} + \frac{3}{5\sqrt{5}} \\ &= \frac{11}{5\sqrt{5}} \quad \text{or} \quad \frac{11\sqrt{5}}{25} \end{aligned}$$

$$\text{e. } \log_b a = 2 \Rightarrow a = b^2 \quad \text{--- ①}$$

$$\log_c b = 3 \Rightarrow b = c^3 \quad \text{--- ②}$$

sub ② into ①

$$a = (c^3)^2$$

$$a = c^6$$

substitute.

$$\text{let } \log_a c = x$$

$$\Rightarrow c = a^x$$

$$c = (c^6)^x$$

$$c^1 = c^{6x}$$

$$\therefore 6x = \frac{1}{x} \quad \therefore x = \frac{1}{6}$$

$$\therefore \log_a c = \frac{1}{6}$$